LAHORE UNIVERSITY OF MANAGEMENT SCIENCES
Department of Electrical Engineering

AI 501 Mathematics for Artificial Intelligence
Quiz 2 Solutions

Name: _____

Campus ID: _____

Total Marks: 10
Time Duration: 20 minutes

Question 1 (3 marks)

Consider the following **training set** of 6 points in 2D (x_1, x_2), with binary labels {Red, Blue}:

Point	**(x_1, x_2)**	**Label**
A	(1, 7)	Red
B	(2, 3)	Red
C	(3, 5)	Blue
D	(5, 2)	Blue
E	(6, 6)	Blue
F	(7, 1)	Red
G	(8, 4)	Red

Predict the label of the **test point** $(4, 4)$ using $k = 3$ and Manhattan distance.

Solution: First, we compute the distance using Manhattan distance as

$$d_M(A) = |4-1| + |4-7| = 6 \quad \text{(Red)}$$
$$d_M(B) = |4-2| + |4-3| = 3 \quad \text{(Red)}$$
$$d_M(C) = |4-3| + |4-5| = 2 \quad \text{(Blue)}$$
$$d_M(D) = |4-5| + |4-2| = 3 \quad \text{(Blue)}$$
$$d_M(E) = |4-6| + |4-6| = 4 \quad \text{(Blue)}$$
$$d_M(F) = |4-7| + |4-1| = 6 \quad \text{(Red)}$$
$$d_M(G) = |4-8| + |4-4| = 4 \quad \text{(Red)}$$

Sorted: $C(2, B),\ B(3, R),\ D(3, B),\ E(4, B),\ G(4, R),\ A(6, R),\ F(6, R)$.

Neighbors $\{C(B), B(R), D(B)\} \Rightarrow$ Majority = $\boxed{\text{Blue}}$.

Question 2 (2 marks)

What will be the dimensions of D in the following scenarios?

(a) If $ABD \in \mathbb{R}^{m \times m}$, where $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. [1 mark]

Solution: To find the dimensions of D, we first determine the dimensions of AB. The product AB will have dimensions $m \times p$ since the inner dimensions n match. Now, for ABD to be $m \times m$, D must have dimensions $p \times m$ to allow for multiplication with AB. Thus, the dimensions of D will be $p \times m$.

(b) If $XD\mathbf{x} \in \mathbb{R}^{d \times 1}$, where $\mathbf{x} \in \mathbb{R}^{d \times 1}$ and $X \in \mathbb{R}^{d \times n}$. [1 mark]

Solution: To find the dimensions of D, notice that X has dimensions $d \times n$. For XD to be defined, the number of columns in X must match the number of rows in D. Therefore, D must have dimensions $n \times k$ for some integer k. Next, for the product $XD\mathbf{x}$ to result in a $d \times 1$ vector, product $(XD)\mathbf{x}$ is as multiplication between a $n \times k$ matrix and a $d \times 1$ vector. This means that k must be equal to d. Thus, the dimensions of D will be $n \times d$.

Question 3 (3 marks)

Suppose a matrix $C \in \mathbb{R}^{n \times d}$ such that its rows consist only of the **basis vectors** of a feature space in $\mathbb{R}^d$. If a vector $\mathbf{v} \in \mathbb{R}^d$ is represented in this feature space as $\mathbf{v} = C^T\mathbf{x}$, answer the following:

(a) The dimensions of C^T. [1 mark]

Solution: The dimensions of C^T will be $d \times n$. This is because the transpose of a matrix swaps its rows and columns.

(b) The dimensions of $\mathbf{x}$. [1 mark]

Solution: The dimensions of $\mathbf{x}$ must be $n \times 1$. This is because C is an $n \times d$ matrix, and for the matrix-vector multiplication $C^T\mathbf{x}$ to be defined, $\mathbf{x}$ must have n rows. The resulting vector $\mathbf{v}$ will have dimensions $d \times 1$.

(c) What does the span of the rows of C represent in this context? [1 mark]

Solution: The span of the rows of C represents the feature space in $\mathbb{R}^d$. Since the rows of C are basis vectors, their span forms a subspace of $\mathbb{R}^d$ that can be used to represent any vector in that space through linear combinations of these basis vectors.

Question 4 (2 marks)

Given a matrix A and a vector $\mathbf{b}$ as follows:

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 4 & -2 \\ 0 & 5 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 7 \\ 3 \\ 8 \end{pmatrix}$$

Compute the product $A\mathbf{b}$ as a linear combination of the columns of A.

Solution: To compute the product $A\mathbf{b}$, as a linear combination of the columns of A:

$$A\mathbf{b} = 7\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + 3\begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix} + 8\begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$$